Hieu Hoang

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Aerospace Engineer | Propulsion, Thermal-Fluid Analysis, Spacecraft Systems, Engineering Automation

SUMMARY

Aerospace engineering graduate focused on propulsion systems, spacecraft hardware, thermal-fluid analysis, experimental data reduction, CAD/CAE, and engineering computation. Led a CubeSat-class resistojet upgrade project involving R134a cold-gas baseline analysis, gas-side heat-transfer modeling, heater sizing, and future vacuum-test planning. Experienced with Python, MATLAB, C++, Arduino, LaTeX, Excel-based engineering analysis, additive manufacturing screening, internal engineering web tools, and local AI-assisted engineering workflows.

EDUCATION

Missouri University of Science and Technology

B.S. Aerospace Engineering, May 2025 | GPA: 3.15/4.00

Focus: propulsion, compressible flow, gas dynamics, thermal-fluid systems, aerospace design, experimental methods, optimization, spacecraft systems, CAD/CAE, engineering computation.

John Wood Community College

A.S. Engineering, 2021 | Co-founder, Engineering Club

CORE STRENGTHS

- Propulsion and spacecraft systems
- Thermal-fluid modeling and test planning
- Python/MATLAB engineering computation
- Experimental data reduction
- CAD/CAE and technical documentation
- AI-assisted engineering automation

EXPERIENCE AND LEADERSHIP

Missouri S&T Satellite Research Team (M-SAT) - Propulsion Research Team Five Lead / Member — Aug. 2022 - May 2025

- Led a Spring 2025 propulsion research effort to upgrade an existing CubeSat-class R134a cold-gas thruster setup into a warm-gas resistojet concept.
- Developed gas-side heating, internal convection, cylindrical conduction, and nichrome heater-wire sizing calculations for a compact resistojet-style flow path.
- Estimated approximately 35.64 W of heating power and a required inner wall temperature near 380-385 deg C for the target gas temperature case.
- Analyzed prior vacuum-chamber cold-gas thrust data and identified likely non-linear scaling causes including distributor pressure drop, flow starvation, turbulence, and pressure recovery limits.

Vinnotek - Engineering Analysis, Additive Manufacturing, and AI Automation Projects — 2026

- Built the Cold Plate Master Baseline Viewer workflow for copper-AM GPU cold-plate design, connecting validated Python solvers to an internal React/TypeScript review interface with 3D geometry viewing, candidate comparison, KPI stackups, optimizer, report export, and manufacturability status.
- Documented the reusable master baseline framework for wavy fins, straight fins, pin fins, TPMS/gyroid, lattice, and hybrid core studies while separating geometry-ranking metrics from full junction-to-coolant claim logic.
- Added coverage-ratio checks, sensitivity cases, CSV/JSON/Markdown reporting, and validation gates for supplier coupon testing, CFD/CHT, roughness, open-channel area, and 575 W thermal margin review.
- Designed a local AI pre-sales automation workflow using Ollama/qwen3:14b, n8n, PowerShell, JSON pipelines, and Excel scoring contracts to rank industrial AM prospects with human review before CRM handoff.

CR Metals Inc. - Entry-Level Engineering Support — St. Louis, MO

- Supported practical engineering and manufacturing tasks requiring technical communication, process understanding, and hands-on problem solving.

SELECTED ENGINEERING PROJECTS

Resistojet Nozzle Thruster Upgrade

Modeled a warm-gas resistojet upgrade for an R134a cold-gas CubeSat propulsion test system; linked thrust baseline data, exhaust velocity targets, heat-addition requirements, material recommendations, and future vacuum-test planning.

Shock Tube Fanno Flow Analysis

Processed pressure transducer voltage data using calibration, smoothing, pressure ratios, normal-shock relations, and Fanno-flow equations while documenting timing and reflected-shock uncertainty.

Wind Tunnel Wing Aerodynamics Analysis

Computed lift coefficient, drag coefficient, moment coefficient, and drag polar behavior from wind-tunnel force and moment data across angle-of-attack sweeps.

Cold Plate Master Baseline Viewer

Built an internal engineering viewer for GPU cold-plate baseline review, including live candidate comparison, junction-to-coolant KPI tracking, manufacturability status, 3D geometry viewing, optimizer support, and report export.

SR-30 Gas Turbine Performance Analysis

Analyzed turbojet lab data at 50,000 and 70,000 RPM to estimate thrust, compressor work, turbine work, fuel-flow behavior, and thermal efficiency with data-quality caveats.

TECHNICAL SKILLS

Aerospace: propulsion, cold-gas thrusters, resistojet thermal modeling, compressible flow, Fanno flow, gas turbine analysis, turbofan cycle modeling, aerodynamics, heat transfer, thermal resistance networks.

Programming: Python, NumPy, Matplotlib, MATLAB, Octave, C++, C, Arduino, PowerShell, Node.js, TypeScript, React, Three.js, Excel engineering computation, Monte Carlo, RSM, root solving.

CAD and tools: FreeCAD, CQ-Editor, Fusion 360 familiarity, LaTeX, Overleaf, AIAA formatting, Git, VS Code, technical documentation.

Testing and AM: Arduino MEGA 2560, solenoid valve control, pressure transducers, vacuum-chamber planning, wind-tunnel data, gas turbine lab data, copper AM screening, FDM/FFF, continuous-fiber FDM, DfAM.